

Pelagic — Technical Supplement

Mani Marami Milani

2026 · Local publication-review edition

S0. Identity and evidence rules

This supplement accompanies the independent Pelagic technical preprint. Primary runtime: `bce3571b0300ecfe5dc6e5dd45f28a9cda57006e`. Later documentation commits do not change the runtime. Asset2 `99ac8336be832c18230dcadefd815308c6c83c8f` remains an unapproved experiment. No release, DOI, arXiv submission or peer review is implied.

Three evidence types are kept separate: source-derived diagrams, actual browser captures, and capture-free performance measurements. The diagrams explain code; they are not screenshots. Video preserves observed browser timing and records frame timestamps before encoding. A 30 fps file does not prove the application rendered 30 unique frames each second. Benchmark data is collected without screencasting, screenshots or concurrent video encoding.

S1–S12. Motion catalog

ID	Subject and claim	Inspection task
S1	Approved pulse in specimen view	Follow several contraction/refill/coast cycles, not one favorable silhouette
S2	Stateful appendage motion	Compare body turn with lagging membrane/tentacle history; inspect underside
S3	Connected activation	Observe animal response, nearby particulate and decay; do not call it a fluid solve
S4	Population during camera movement	Watch changing projected scale and detail, retaining original behavior
S5	Bubble passage	Look for moving background distortion through a hero, not merely its outline
S6	Four authored camera views and Explore	One world persists through observer changes

ID	Subject and claim	Inspection task
S7	Sanctuary plume	Inspect localized discharge, sparse ecology and dark geological scale
S8	Idle condensation	Ocean stays alive while the clock forms in the same compositor
S9	Contact and merging at normal size	Surface approach, narrow neck and transfer must be visible without a diagnostic zoom
S10	Pointer deformation and recovery	Persistent field remembers input and settles afterward
S11	Minute change	Changed cells retract and reconnect; unchanged digits retain their place
S12	Pinch dismissal	Throat narrows and separates before optical presentation disappears

Every clip has a JSON record containing SHA, backend, browser, viewport, drawing buffer, DPR, camera state, seed, simulation time, duration, observed source rate, encoded rate and capture tool path. Master videos and frame timestamps are staged outside normal Git. The small committed gallery and manifest point to local-review assets until a future release is actually authorized. No invented remote video URL is used.

S13. Numerical and timing interpretation

The paper’s fixed-step values belong to different accumulators, not one global physics solve. Tissue uses 1/60 s with bounded admission; the journey follower uses 1/240 s; idle admits at most three 1/60 s steps per update. Main-route locomotion uses its own bounded delta and exponential factors. Therefore the whole application is not bitwise frame-rate independent merely because several subsystems use fixed steps.

The current field’s analytical ambient expression is divergence-free before local effects. Its envelopes, drift and speed clamp invalidate a general incompressibility claim. The word “wake” names a visual disturbance seeded by motion/pulse observations, not a solved vorticity field. The secondary thrust window supplies a perceptual rhythm inspired by biology without estimating real pressure or efficiency.

The liquid field’s velocity clamp precedes the queued event impulse. Its final displacement remains clamped, and edges are attenuated. Reporting only the first clamp as a bound on every intermediate would be inaccurate. Similarly, the implicit field normal uses differences at a 1.7-pixel spacing without dividing by that spacing. It is an authored optical normal, not a numerically convergent measurement of a physical height derivative.

S14. Historical negative results

These are development records, not controlled comparisons against other research systems. The evidence links point to the relevant versioned records; their original local-review URLs are not reproduction dependencies.

N1. Camera ownership

Hypothesis: independent camera position, point-of-interest and safety corrections could keep moving animals attractively framed. **Implementation:** earlier camera experiments combined multiple correction sources. **Observed failure:** the author rejected overdirected/janky travel. **Replacement:** bounded input progress and baked position/quaternion tracks. **Evidence:** the M5/M5.2 history and current `JourneyController/CameraTrack`. **Conclusion boundary:** this explains an ownership decision; no unsupported jerk reduction or isolated performance percentage is published.

N2. Giant analytic lens

Hypothesis: one large analytic lens would clearly demonstrate live refraction. **Implementation:** the M3 development lens used a world-space ellipsoid and the real scene target. **Observed failure:** technical bending was accepted before the large dark-disc presentation was artistically accepted. **Replacement:** a brief passage of sparse refractive hero bubbles with cheaper ambient films. **Evidence:** M3 review history and `BubblePopulation`. **Conclusion boundary:** the optical architecture survived; rejecting the presentation does not mean analytic refraction failed.

N3. Bright repetitive geology

Hypothesis: larger shelf masses and broad color zones would clarify the abyss. **Observed failure:** successive M6 reviews found flat, bright or repetitive slabs. **Replacement:** connected meso geology, darker mineral response, restrained highlights and deterministic silhouette variation. **Evidence:** M6.4–M6.6 reports and `shelfSilhouettes/materials`. **Conclusion boundary:** the result is an art-directed basalt garden, not a photogrammetric geological reconstruction or physically measured material.

N4. Separate idle renderer

Hypothesis: a transparent independent idle canvas could supply a liquid clock over a moving ocean. **Observed limitation:** that canvas did not contain the live ocean color to refract. **Replacement:** M7 integrates topology and field deformation into the existing live-ocean optical output. **Evidence:** historical `IdleGlassScene.jsx`, current `IdleScreen.jsx`, `OceanIdleGlass` and `LiveOceanLens`. **Conclusion boundary:** the old file is retained history, not the active architecture; no second ocean is maintained.

N5. Invalid pointer versus idle corruption

Initial hypothesis: idle return damaged dynamic geometry. **Investigation:** strict QA identified malformed synthetic pointer events lacking finite spatial coordinates before the symptom appeared.

Correction: finite spatial-input validation at the boundary, not sanitizing final geometry. **Evidence:** M6.6.1 input/idle-return test history and guarded pointer path. **Conclusion boundary:** this does not prove every context-loss or frozen-tab case is fixed. It demonstrates why the first non-finite input matters more than the eventual renderer symptom.

N6. Asset2 / optional GTAO — not the publication runtime

Hypothesis: material grounding and a narrowly scoped GTAO experiment could improve a background reference match without changing approved subjects. **Implementation:** a review-only environment pass and a separate QA-only AO prototype using the pinned scene depth. **Observed failure:** the first shader adapter had a depth-fetch type mismatch; the corrected prototype compiled but returned identity AO. Depth readback showed no useful variation. The installed WebGL multisample fallback resolves color with COLOR_BUFFER_BIT only.

Evidence: Asset2 GTAO_PROTOTYPE.md at documentation commit:

b4ca42faa3b80e465d39995ebb0537a2aa948148

Its retained range/readback record documents the failed depth experiment. The main image-metric improvement was about 1.47%, not the requested 30%. Pooled phases improved only about 0.55%, with an adverse phase retained. **Replacement:** the candidate retained material-space approximation, not GTAO. **Conclusion boundary:** those timings do not measure useful AO. Asset2 is not approved and is excluded from primary publication frames and fresh results.

S15. Optical information boundary

The clean ocean color is real and current. This establishes live image distortion, not complete ray transport. The following limits remain:

- Transparent tissue generally writes no depth; its visible color is already composited with other layers.
- Opaque-depth masking is coded, but useful depth resolution is backend- dependent. The pinned NVIDIA WebGL MSAA issue is not repaired by publication.
- A bounded virtual image plane supplies missing transparent-layer depth.
- Offscreen color cannot be recovered; border fades and displacement caps reduce artifacts rather than reconstruct it.
- Overlapping bubble/thermal/idle contributions sample one clean source. They do not trace nested refractions through each other's outputs.
- Idle thickness is a scalar field with an artistic normal. It is not a watertight three-dimensional liquid volume.

S16. Reproduction and fresh results

Follow the root reproducibility guide, lockfile and publication manifest. Build before running the packaging test. Benchmark tools record raw frame intervals and summary statistics; they never rename those intervals GPU time. Tables in the primary paper are derived from `results/benchmark.json`. Historical milestone numbers remain separate.

Browser-generated desktop/narrow/portrait input is not physical device testing. The publication does not claim Safari, Firefox, hardware WebGPU or mobile validation absent a named record. The legacy full-ocean WebGPU problem remains outside this publication task. Existing runtime source is held byte-identical.

S17. Licensing and authorship

Author: **Mani Marami Milani**. Codex/OpenAI assistance covered implementation, audit, writing and tooling under the author's direction. Original code is MIT; original paper, diagrams, documentation and project-generated media are CC BY 4.0. Third-party exceptions remain controlling and are enumerated in asset provenance and retained notices. External reference artwork is not copied into this supplement. Preparation of a source archive does not assert arXiv submission, DOI assignment, acceptance or peer review.